Chicago Crash Cause Prediction

Machine Learning Project for Traffic Safety Insights

Group 2: Presentation

Project Overview

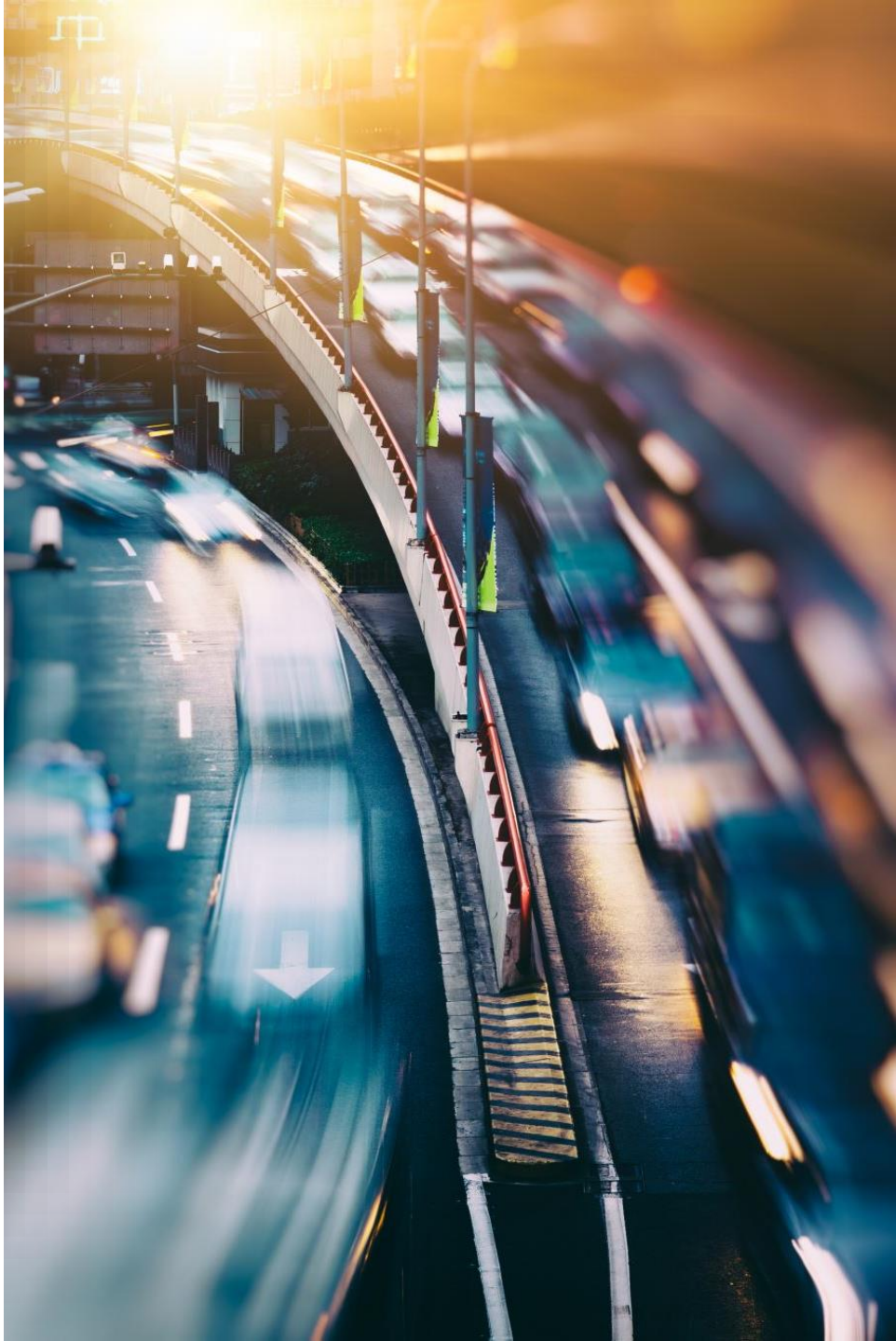
- Predicting primary contributory causes of traffic crashes in Chicago using machine learning.
- Goal: Improve road safety through data-driven insights.

Business Understanding

- Chicago experiences rising traffic accidents.
- Data is biased due to post-crash information.
- Objective: Build predictive model for crash causes.

Stakeholders

- - City Planners: Infrastructure improvements
- - Traffic Safety Authorities: Safety campaigns & enforcement
- - Policy Makers: Data-driven regulations



Objectives

- Build classification model for crash causes
- Identify key contributing factors
- Provide actionable insights to reduce accidents

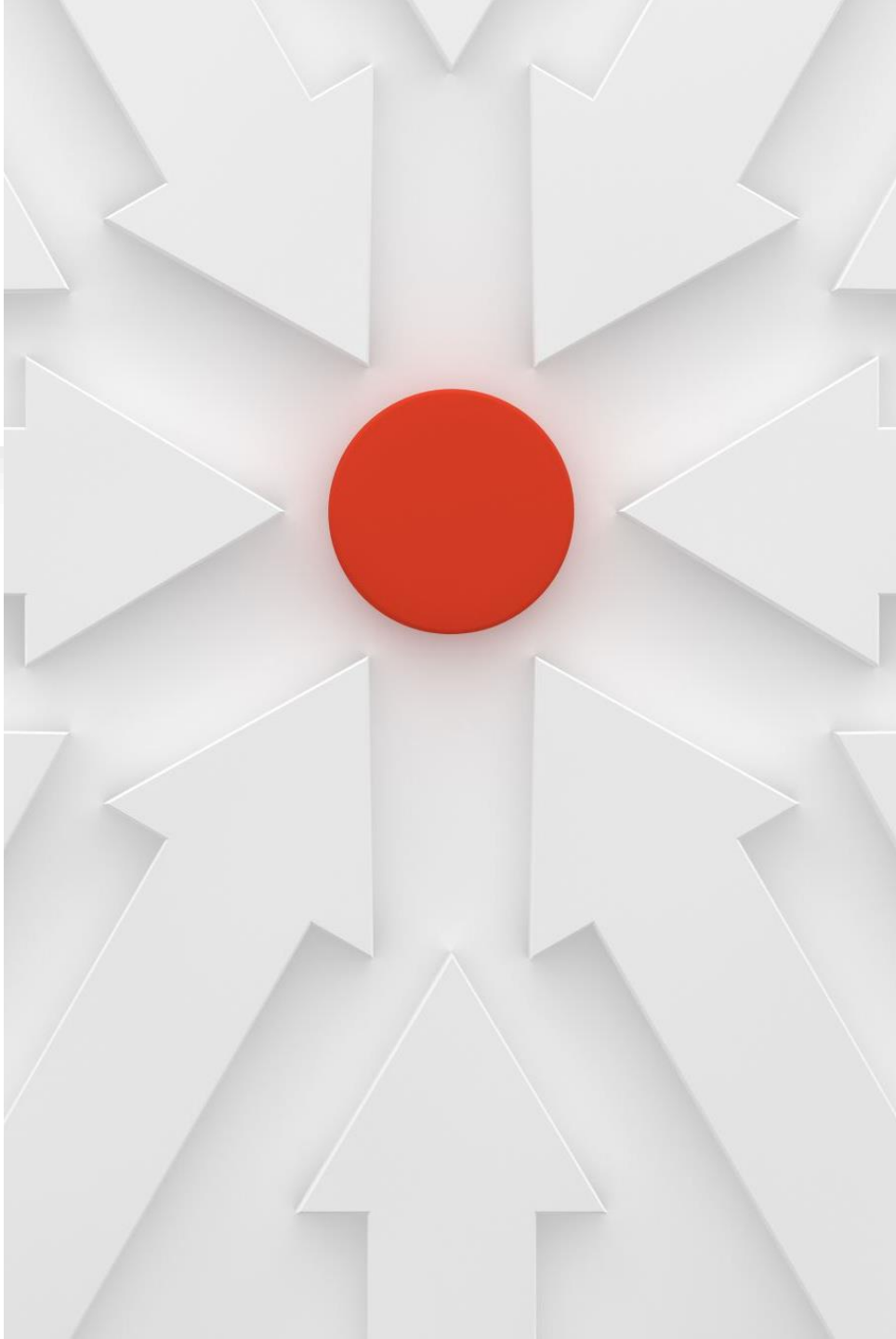
Data Sources

Chicago Data Portal datasets:

- Traffic Crashes
- Vehicles
- People
- Merged into unified dataset for analysis.

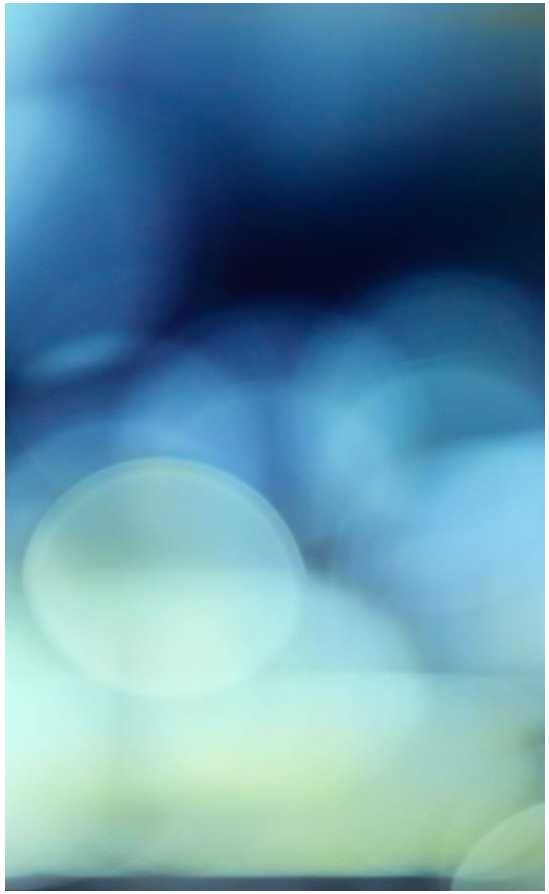
Target Variable

- PRIM_CONTRIBUTORY_CAUSE
- Multi-class classification problem representing main crash cause.



Project Flow

- Business Understanding
- Data Understanding
- Data Processing
- Feature Engineering
- Pipeline Development
- Modeling
- Evaluation
- Deployment

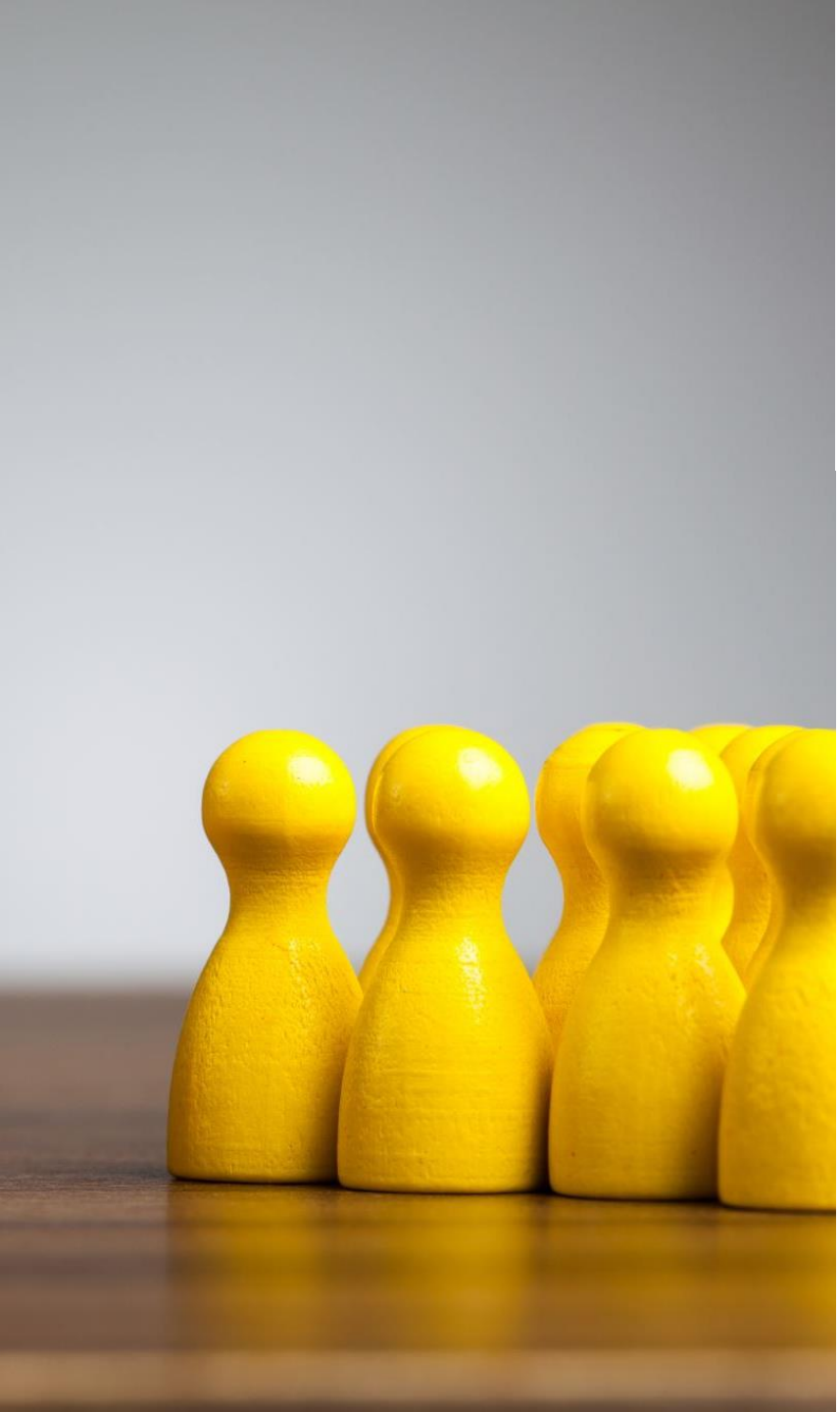


Feature Engineering

- Selection of relevant predictors
- Removal of high-cardinality features
- Avoidance of overfitting
- Handling categorical and numerical variables

Machine Learning Pipeline

- Train-test split
- Data cleaning
- Encoding categorical variables
- Normalization of numerical features

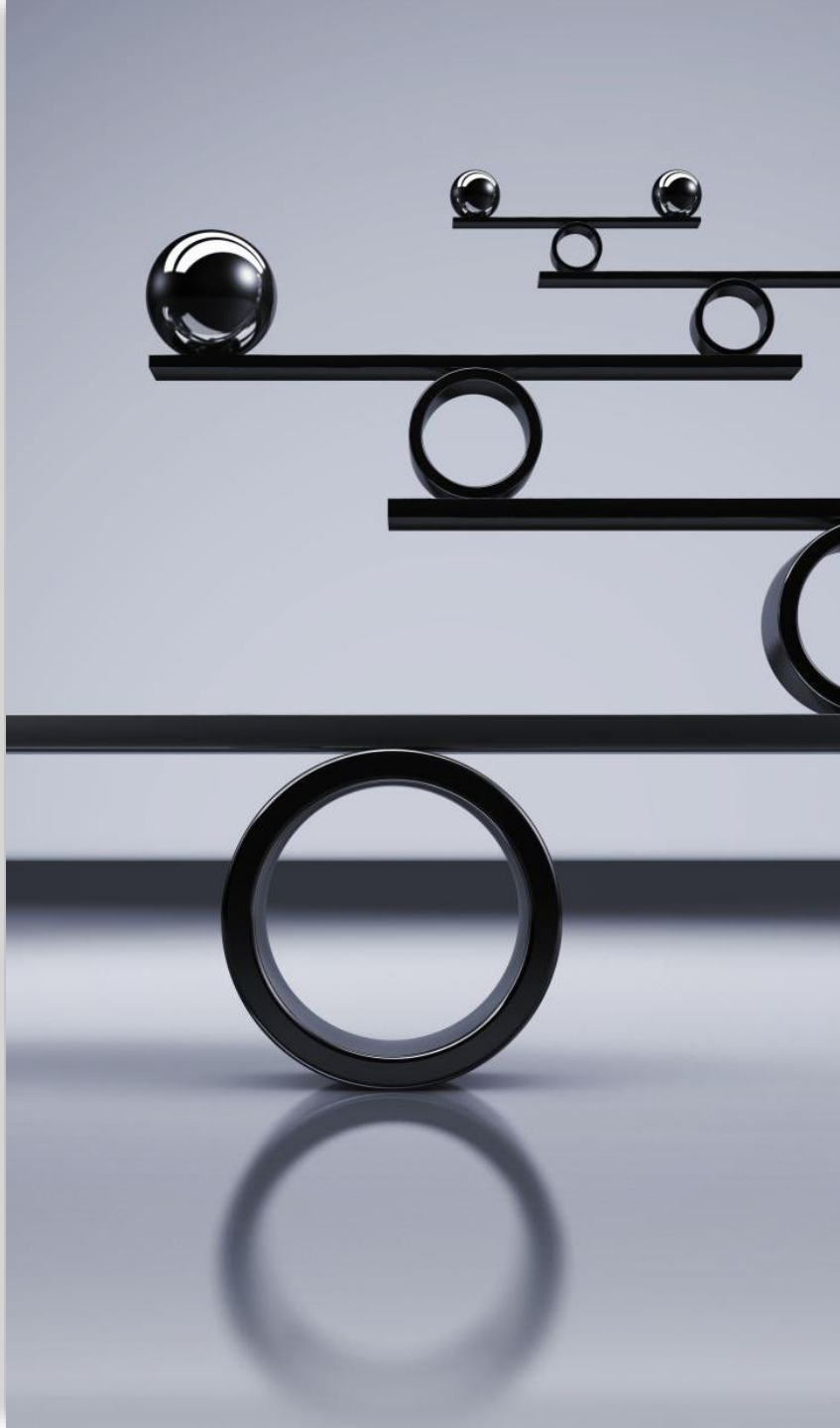


Modeling Approach

- Multiple classification models trained
- Comparison of performance across models
- Focus on generalization and robustness

Evaluation Metrics

- Macro F1-score (handles imbalance)
- Confusion matrix analysis
- Feature importance (SHAP)
- Baseline comparison



Key Challenges

IMBALANCED CLASSES

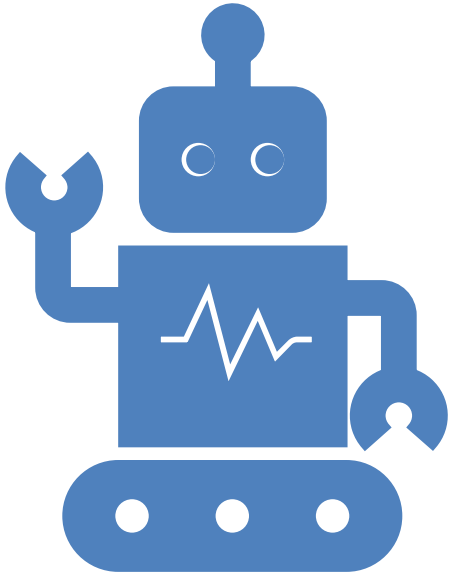
HIGH CARDINALITY FEATURES

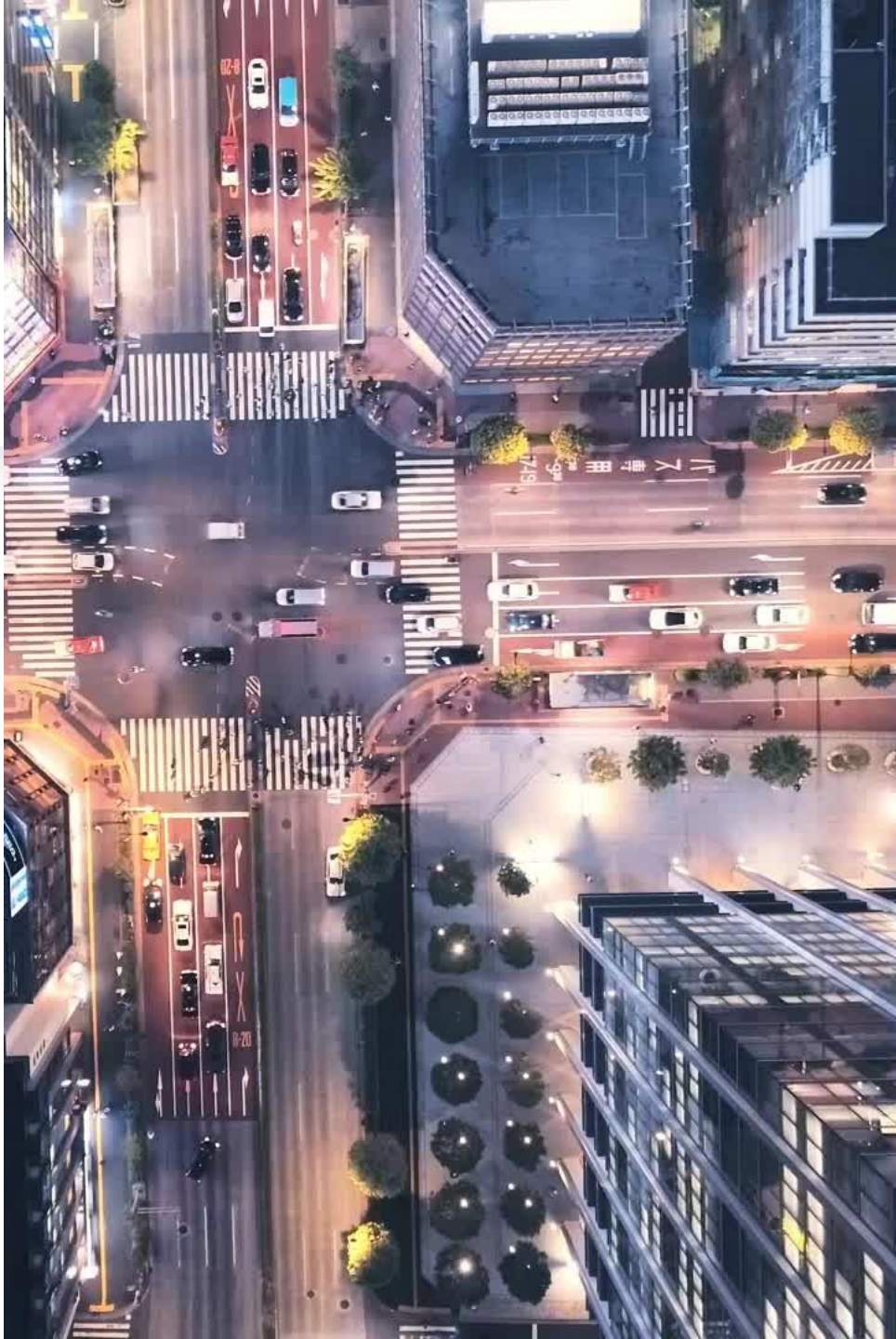
MULTI-TABLE DATASET INTEGRATION

DATA LEAKAGE RISKS (POST-CRASH VARIABLES)

Deployment

- Model serialized using Joblib
- Designed for stakeholder use
- Enables real-time prediction of crash causes



Conclusion

- This project enables predictive insights into traffic crash causes.
- Supports safer city planning and policy decisions.